ME403 – Introduction to Robotics

Lab 00: Introduction Week Handout

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Lab Outcomes

By the end of this lab, you will be able to:

- Set up a Python virtual environment for robotic control.
- Connect to the Dobot Magician over USB-serial using the GUI.
- Connect to the DOBOT MG400 over Ethernet and verify network setup.
- Move individual joints and understand each joint's role and limits.
- Observe how joint angles map to the end-effector position using live pose readout.

1. Overview

Both the **Dobot Magician** (USB-serial) and the **DOBOT MG400** (Ethernet/TCP-IP) are covered in this lab. The two software tracks are organized under `magician/` and `mg400/` folders inside `Students/00_IntroductionWeek/`.

Provided Files

Magician (`magician/`):

File	Purpose
`00_magician_gui.py`	GUI application: port discovery, connection, alarm clearing, joint control, live pose readout.
`utils.py`	Shared helper functions (safe bounds, port finder, move wrappers).
`requirements.txt`	Python dependencies (install with `pip`).

MG400 (`mg400/`):

File	Purpose
`00_mg400_gui.py`	GUI application: robot selector, TCP connection, error clearing, joint control, live pose readout.
`utils_mg400.py`	Shared helper functions (safe bounds, TCP connect, move wrappers, error handler).
`requirements.txt`	Python dependencies (install with `pip`).

2. Quick Start Guide

> **Note: Avoid PDF Copy-Paste for Commands**
> Type these commands manually or copy from the digital version of this document. Some PDF viewers insert extra spaces in folder names (e.g. `. venv` instead of `.venv`), which will cause errors.

Magician Setup

Magician — Environment Setup

Run from: `Students/00_IntroductionWeek/`

Linux:

```
# Run from: Students/00_IntroductionWeek/
python3 -m venv .venv
source .venv/bin/activate
pip install -r magician/requirements.txt
cd magician
python 00_magician_gui.py
```

Windows (PowerShell):

```
# Run from: Students\00_IntroductionWeek\
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r magician\requirements.txt
cd magician
python 00_magician_gui.py
```

MG400 Setup

MG400 — Environment Setup

Step 1 — Network:

- Check the sticker on your assigned MG400 and note its **Robot Number** and **IP Address**.
- Set your PC's Ethernet adapter to static IP `192.168.2.100`, subnet `255.255.255.0`.
- Verify connectivity by pinging your assigned robot IP from the **Robot Network Table** (e.g., `ping 192.168.2.7`).

Step 2 — SDK clone (one-time, run from `dobot_ws/` root):

```
git clone https://github.com/Dobot-Arm/TCP-IP-4Axis-Python.git vendor/
    TCP-IP-4Axis-Python
```

Step 3 — Install and run:
Linux:

```
# Run from: Students/00_IntroductionWeek/
python3 -m venv .venv
source .venv/bin/activate
pip install -r mg400/requirements.txt
cd mg400
python 00_mg400_gui.py
```

Windows (PowerShell):

```
# Run from: Students\00_IntroductionWeek\
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r mg400\requirements.txt
cd mg400
python 00_mg400_gui.py
```

3. Hardware Setup

Dobot Magician

1. **Surface:** Place on a flat, stable surface. Clear a 40 cm radius around the base.
2. **Posture:** Manually set the arm to a neutral posture (both segments at ≈45°) before powering on.
3. **Power:** Connect the 12 V DC adapter. **USB power alone is insufficient.**
4. **Communication:** Connect the USB cable (CP210x USB-to-UART bridge).

Status Signals:

- **Green LED + Single Beep:** Ready.
- **Red LED + Rapid Beeping:** Alarm state — click **Clear Alarms** in the GUI.
- **Yellow/Flashing LED:** Busy (homing or executing a command).

DOBOT MG400

1. **Surface:** Place on a flat, stable surface. Clear a 50 cm radius around the base.
2. **Power:** Connect the power adapter and press the power button.
3. **Communication:** Connect an Ethernet cable between the robot and your PC.
4. **Network:** Use the same network settings from **MG400 Setup, Step 1**. Verify by pinging the IP shown on your robot sticker (see Robot Network Table).

4. OS-Specific Notes

4.1. Linux (Magician only)

Users must have permission to access serial devices. Run the following once:

```
sudo usermod -a -G dialout $USER
# Log out and log back in for changes to take effect.
```

The Magician typically appears as `/dev/ttyUSB0` or `/dev/ttyACM0`.

4.2. Windows

The Magician driver is usually handled by Windows Update. To find the port manually:

1. Open **Device Manager**.
2. Expand **Ports (COM & LPT)**.

3. Look for **Silicon Labs CP210x USB to UART Bridge** (e.g., `COM3`).

The GUI auto-discovers the port, so manual lookup is typically unnecessary.

5. The Dobot Magician

The **Dobot Magician** is a 4-axis desktop robotic arm with:

- **Reach:** ≈320 mm maximum.
- **Payload:** 500 g at full extension.
- **Repeatability:** ±0.2 mm.
- **Communication:** USB-serial (CP210x chip).

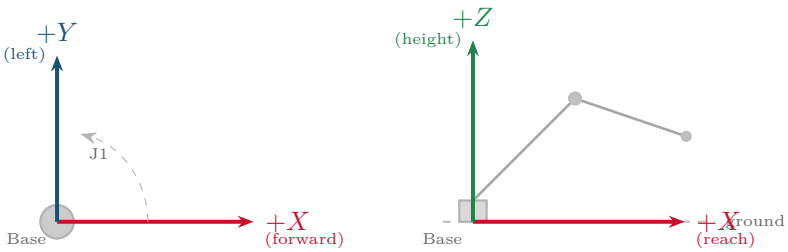
The arm has two main segments connected by revolute joints:

- **Upper arm (L1):** 135 mm — from the shoulder (J2) to the elbow.
- **Forearm (L2):** 147 mm — from the elbow to the end-effector.

A **parallel linkage** on J3 keeps the forearm's firmware angle relative to the upper arm (body-frame), rather than world-frame absolute.

5.1. Coordinate System

The two diagrams below show the Magician coordinate system from two viewpoints.



Top View (X–Y) **Side View (X–Z)**

- **X-axis:** Forward (pointing away from the base front face).
- **Y-axis:** Left (positive to the robot's left when facing forward).
- **Z-axis:** Vertical upward.
- **Units:** Millimeters (mm).
- **Rotation (R):** Tool rotation in degrees (°).

5.2. Joint Definitions and Limits

Each joint controls a different degree of freedom. The GUI enforces these limits automatically (values are clamped before sending to the robot).

Joint	Motion	Safe Range	Units
J1	Base Rotation	−90 to 90°	deg
J2	Rear Arm Elevation	0 to 85°	deg
J3	Forearm Elevation	−10 to 85°	deg
J4	End-Effector Rotation	−90 to 90°	deg

Cartesian Dimension	Safe Operating Range	Unit
X (Reach)	120 to 315 mm	mm
Y (Slew)	±158 mm	mm
Z (Height)	5 to 155 mm	mm

6. The DOBOT MG400

The **DOBOT MG400** is a 4-axis collaborative robotic arm designed for flexible benchtop use:

- **Reach:** 440 mm maximum.
- **Payload:** 500 g.
- **Repeatability:** ±0.02 mm.
- **Communication:** Ethernet/TCP-IP (three ports: dashboard, motion, feedback).

Key differences from the Magician: larger reach, and **Z cannot be negative** — the origin (Z = 0) is the robot's mounting surface. Always keep Z ≥ 5 mm to avoid collision.

6.1. Coordinate Bounds

Axis	Safe Range	Notes	Unit
X	60 – 400	Inner ≈60 mm (base singularity); max reach 440 mm	mm
Y	±220	Symmetric lateral sweep	mm
Z	5 – 140	**Cannot go negative** — 0 = mounting surface	mm
R	±170	End-effector rotation	deg

6.2. Joint Limits

Joint	Motion	Safe Range	Notes
J1	Base Rotation	±160°	
J2	Shoulder Elevation	−25 to 85°	
J3	Elbow	−25 to 105°	Firmware absolute = J2 + J3_body
J4	Wrist Rotation	±180°	

6.3. Robot Network Table

Robot Number	IP Address	Notes
Robot 1	192.168.2.7	Match with sticker on the robot
Robot 2	192.168.2.10	Match with sticker on the robot
Robot 3	192.168.2.9	Match with sticker on the robot
Robot 4	192.168.2.6	Match with sticker on the robot

6.4. Using the MG400 GUI

1. **Launch:** `cd mg400 && python 00_mg400_gui.py`
2. **Connect:** Select the robot number that matches the sticker on your assigned arm (Robot 1–4), then click **Connect**. The GUI enables the robot and moves to READY_POSE automatically.
3. **Move Joints:** Use the spinboxes and +/− buttons to set target joint angles, then click **Move** for one joint, or **Send All** for all four.
4. **Live Pose:** The right panel shows current X/Y/Z/R and J1–J4, updated every 500 ms.
5. **Joint Limits:** The limits panel shows the safe range for each joint.
6. **Home:** Returns to READY_POSE (300, 0, 50, 0) mm.
7. **ESTOP:** Sends EmergencyStop immediately and disconnects.

7. Using the Magician GUI

1. **Launch:** `cd magician && python 00_magician_gui.py`
2. **Connect:** Select the port (auto-discovered with a star marker) and click **Connect**. The GUI clears any alarms and homes the robot automatically.
3. **Move Joints:** Use the spinboxes and +/− buttons to set a target angle, then click **Move** for that joint, or **Send All** to move all four joints at once.
4. **Live Pose:** The right panel shows the current X, Y, Z, R and J1–J4 values, updated every 500 ms.
5. **Joint Limits:** The limits panel shows the safe range for each joint.
6. **Details:** The log panel at the bottom shows connection events, alarms, and move commands.

8. Troubleshooting

Common Magician Connection Issues

Symptom	Recommended Action
`Access Denied`	Ensure **DobotStudio** and other Python scripts are closed.
`Device not found`	Verify both USB and Power cables. Click **Refresh** in the GUI to refresh ports.
Motors don't move	Check the 12 V supply. If LED is red, click **Clear Alarms** in the GUI.
Erratic movement	Reset to neutral posture and power cycle the robot.

MG400 Connection Issues

Symptom	Recommended Action
`Connection refused`	Check static IP (`192.168.2.100`), ping your assigned robot IP from the Robot Network Table, and verify the Ethernet cable.
`dobot_api not found`	Clone the SDK at the `dobot_ws/` root (see Quick Start, Step 2).
Robot doesn't move after connect	GUI enables automatically; click **Clear Errors** if the status badge shows ERROR.
Wrong robot moves	Verify the sticker robot number/IP and select the matching robot in the dropdown before clicking Connect.

9. Safety Mandates

Mandatory Safety Rules

1. **Workspace:** Maintain a clean area. No cables in the reach zone (40 cm radius for Magician, 50 cm for MG400).
2. **Initialization:** Always let the GUI complete its connection sequence (alarm clear + homing) before sending commands.
3. **Joint Limits:** The GUI clamps all values to the safe ranges shown in the Joint Limits panel.
4. **MG400 Reach:** The MG400 arm extends to 440 mm — maintain a 50 cm clear radius around the base.
5. **Supervision:** Stay within reach of the power cable at all times.